

ADAM: A METHOD FOR STOCHASTIC OPTIMIZATION

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ABSTRACT

We introduce *Adam*, an algorithm for first-order gradient-based optimization of stochastic objective functions, based on adaptive estimates of lower-order moments. The method is straightforward to implement, is computationally efficient, has little memory requirements, is invariant to diagonal rescaling of the gradients, and is well suited for problems that are large in terms of data and/or parameters. The method is also appropriate for non-stationary objectives and problems with very noisy and/or sparse gradients. The hyper-parameters have intuitive interpretations and typically require little tuning. Some connections to related algorithms, on which *Adam* was inspired, are discussed. We also analyze the theoretical convergence properties of the algorithm and provide a regret bound on the convergence rate that is comparable to the best known results under the online convex optimization framework. Empirical results demonstrate that *Adam* works well in practice and compares favorably to other stochastic optimization methods. Finally, we discuss *AdaMax*, a variant of *Adam* based on the infinity norm.

1 INTRODUCTION

Stochastic gradient-based optimization is of core practical importance in many fields of science and engineering. Many problems in these fields can be cast as the optimization of some scalar parameterized objective function requiring maximization or minimization with respect to its parameters. If the function is differentiable w.r.t. its parameters, gradient descent is a relatively efficient optimization method, since the computation of first-order partial derivatives w.r.t. all the parameters is of the same computational complexity as just evaluating the function. Often, objective functions are stochastic. For example, many objective functions are composed of a sum of subfunctions evaluated at different subsamples of data; in this case optimization can be made more efficient by taking gradient steps w.r.t. individual subfunctions, i.e. stochastic gradient descent (SGD) or ascent. SGD proved itself as an efficient and effective optimization method that was central in many machine learning success stories, such as recent advances in deep learning (Deng et al., 2013; Krizhevsky et al., 2012; Hinton & Salakhutdinov, 2006; Hinton et al., 2012a; Graves et al., 2013). Objectives may also have other sources of noise than data subsampling, such as dropout (Hinton et al., 2012b) regularization. For all such noisy objectives, efficient stochastic optimization techniques are required. The focus of this paper is on the optimization of stochastic objectives with high-dimensional parameters spaces. In these cases, higher-order optimization methods are ill-suited, and discussion in this paper will be restricted to first-order methods.

We propose *Adam*, a method for efficient stochastic optimization that only requires first-order gradients with little memory requirement. The method computes individual adaptive learning rates for different parameters from estimates of first and second moments of the gradients; the name *Adam* is derived from adaptive moment estimation. Our method is designed to combine the advantages of two recently popular methods: AdaGrad (Duchi et al., 2011), which works well with sparse gradients, and RMSProp (Tieleman & Hinton, 2012), which works well in on-line and non-stationary settings; important connections to these and other stochastic optimization methods are clarified in section 5. Some of *Adam*'s advantages are that the magnitudes of parameter updates are invariant to rescaling of the gradient, its stepsizes are approximately bounded by the stepsize hyperparameter, it does not require a stationary objective, it works with sparse gradients, and it naturally performs a form of step size annealing.

*Equal contribution. Author ordering determined by coin flip over a Google Hangout.

ADAM: 一种随机优化方法

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摘要

我们提出了*Adam*, 这是一种针对随机目标函数的一阶梯度优化算法, 它基于低阶矩的自适应估计。该方法实现简单、计算高效、内存需求小, 对梯度的对角重缩放具有不变性, 非常适合数据和/或参数规模庞大的问题。该方法同样适用于非平稳目标以及梯度噪声很大和/或稀疏梯度的问题。其超参数具有直观的解释, 通常几乎不需要调优。我们还讨论了*Adam* 所借鉴的相关算法之间的联系。此外, 我们分析了该算法的理论收敛性质, 并给出了收敛速率的遗憾界, 该结果可与在线凸优化框架下已知的最佳结果相媲美。实验结果表明, *Adam*在实践中表现良好, 优于其他随机优化方法。最后, 我们讨论了*AdaMax*, 这是一种基于无穷范数的*Adam* 变体。

1 引言

基于随机梯度的优化在科学与工程等诸多领域中具有核心的实践意义。这些领域中的许多问题都可以转化为对某个标量参数化目标函数的优化, 即针对其参数进行最大化或最小化。如果该函数对其参数可微, 那么梯度下降就是一种相对高效的优化方法, 因为计算所有参数的一阶偏导数与仅仅计算函数值具有相同的计算复杂度。目标函数往往是随机的。例如, 许多目标函数由在不同数据子样本上求值的子函数之和构成; 在这种情况下, 通过对单个子函数执行梯度步骤可以使优化更加高效, 即随机梯度下降 (SGD) 或随机梯度上升。SGD 已被证明是一种高效且有效的优化方法, 在许多机器学习的成功案例中处于核心地位, 例如近年来深度学习的进展 (Deng et al., 2013; Krizhevsky et al., 2012; Hinton & Salakhutdinov, 2006; Hinton et al., 2012a; Graves et al., 2013)。除数据子采样外, 目标函数还可能存在其他噪声来源, 例如随机失活 (Hinton et al., 2012b) 正则化。对于所有这些带噪声的目标函数, 都需要高效的随机优化技术。本文的重点是对具有高维参数空间的随机目标函数进行优化。在这些情况下, 高阶优化方法并不适用, 因此本文的讨论将仅限于一阶方法。

我们提出了 *Adam*, 一种高效的随机优化方法, 它只需要一阶梯度, 且内存需求很小。该方法根据梯度的一阶和二阶矩估计, 为不同的参数计算各自的自适应学习率; *Adam*这一名称源自自适应矩估计。我们的方法旨在结合两种近期流行的方法的优点: *AdaGrad* (Duchi et al., 2011), 它在稀疏梯度情况下表现良好; 以及 *RMSProp* (Tieleman & Hinton, 2012), 它在在线和非平稳场景下表现良好; 第 5 节阐明了与这些方法及其他随机优化方法的重要联系。*Adam* 的一些优点包括: 参数更新的幅度对梯度的缩放保持不变, 其步长大致受步长超参数的约束, 它不要求目标是平稳的, 它能处理稀疏梯度, 并且它能自然地执行某种形式的步长退火。

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Algorithm 1: *Adam*, our proposed algorithm for stochastic optimization. See section 2 for details, and for a slightly more efficient (but less clear) order of computation. g_t^2 indicates the elementwise square $g_t \odot g_t$. Good default settings for the tested machine learning problems are $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$. All operations on vectors are element-wise. With β_1^t and β_2^t we denote β_1 and β_2 to the power t .

Require: α : Stepsize

Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates for the moment estimates

Require: $f(\theta)$: Stochastic objective function with parameters θ

Require: θ_0 : Initial parameter vector

$m_0 \leftarrow 0$ (Initialize 1st moment vector)

$v_0 \leftarrow 0$ (Initialize 2nd moment vector)

$t \leftarrow 0$ (Initialize timestep)

while θ_t not converged **do**

$t \leftarrow t + 1$

$g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)

$m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)

$v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ (Update biased second raw moment estimate)

$\hat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ (Compute bias-corrected first moment estimate)

$\hat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ (Compute bias-corrected second raw moment estimate)

$\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$ (Update parameters)

end while

return θ_t (Resulting parameters)

In section 2 we describe the algorithm and the properties of its update rule. Section 3 explains our initialization bias correction technique, and section 4 provides a theoretical analysis of Adam’s convergence in online convex programming. Empirically, our method consistently outperforms other methods for a variety of models and datasets, as shown in section 6. Overall, we show that Adam is a versatile algorithm that scales to large-scale high-dimensional machine learning problems.

2 ALGORITHM

See algorithm 1 for pseudo-code of our proposed algorithm *Adam*. Let $f(\theta)$ be a noisy objective function: a stochastic scalar function that is differentiable w.r.t. parameters θ . We are interested in minimizing the expected value of this function, $\mathbb{E}[f(\theta)]$ w.r.t. its parameters θ . With $f_1(\theta), \dots, f_T(\theta)$ we denote the realisations of the stochastic function at subsequent timesteps 1, ..., T . The stochasticity might come from the evaluation at random subsamples (minibatches) of datapoints, or arise from inherent function noise. With $g_t = \nabla_{\theta} f_t(\theta)$ we denote the gradient, i.e. the vector of partial derivatives of f_t , w.r.t θ evaluated at timestep t .

The algorithm updates exponential moving averages of the gradient (m_t) and the squared gradient (v_t) where the hyper-parameters $\beta_1, \beta_2 \in [0, 1)$ control the exponential decay rates of these moving averages. The moving averages themselves are estimates of the 1st moment (the mean) and the 2nd raw moment (the uncentered variance) of the gradient. However, these moving averages are initialized as (vectors of) 0’s, leading to moment estimates that are biased towards zero, especially during the initial timesteps, and especially when the decay rates are small (i.e. the β s are close to 1). The good news is that this initialization bias can be easily counteracted, resulting in bias-corrected estimates $\hat{m}_t$ and $\hat{v}_t$. See section 3 for more details.

Note that the efficiency of algorithm 1 can, at the expense of clarity, be improved upon by changing the order of computation, e.g. by replacing the last three lines in the loop with the following lines: $\alpha_t = \alpha \cdot \sqrt{1 - \beta_2^t} / (1 - \beta_1^t)$ and $\theta_t \leftarrow \theta_{t-1} - \alpha_t \cdot m_t / (\sqrt{v_t} + \epsilon)$.

2.1 ADAM’S UPDATE RULE

An important property of Adam’s update rule is its careful choice of stepsizes. Assuming $\epsilon = 0$, the effective step taken in parameter space at timestep t is $\Delta_t = \alpha \cdot \hat{m}_t / \sqrt{\hat{v}_t}$. The effective stepsize has two upper bounds: $|\Delta_t| \leq \alpha \cdot (1 - \beta_1) / \sqrt{1 - \beta_2}$ in the case $(1 - \beta_1) > \sqrt{1 - \beta_2}$, and $|\Delta_t| \leq \alpha$

算法 1: *Adam*, 我们提出的随机优化算法。详见第 2 节, 以及一种略微更高效 (但不够清晰) 的计算顺序。 g_t^2 表示逐元素平方 $g_t \odot g_t$ 。对于所测试的机器学习问题, 良好的默认设置为 $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ 和 $\epsilon = 10^{-8}$ 。对向量的所有操作均为逐元素操作。我们用 β_1^t 和 β_2^t 表示 β_1 和 β_2 的 t 次幂。

要求: α : 步长

要求: $\beta_1, \beta_2 \in [0, 1]$: 矩估计的指数衰减率

要求: $f(\theta)$: 含参数 θ 的随机目标函数

要求: θ_0 : 初始参数向量

$m_0 \leftarrow 0$ (初始化 1st 矩向量)

$v_0 \leftarrow 0$ (初始化 2nd 矩向量)

$t \leftarrow 0$ (初始化时间步)

当 θ_t 未收敛时 执行

$t \leftarrow t + 1$

$g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ (在时间步 t 计算随机目标函数的梯度)

$m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (更新有偏一阶矩估计)

$v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ (更新有偏二阶原始矩估计)

$\hat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ (计算经偏差校正的一阶矩估计)

$\hat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ (计算经偏差校正的二阶原始矩估计)

$\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$ (更新参数)

结束循环

返回 θ_t (得到的参数)

在第 2 节中, 我们描述该算法及其更新规则的性质。第 3 节解释我们的初始化偏差校正技术, 第 4 节对 Adam 在在线凸规划中的收敛性进行理论分析。实证结果表明, 如第 6 节所示, 我们的方法在多种模型和数据集上始终优于其他方法。总体而言, 我们表明 Adam 是一种通用算法, 可扩展到大规模高维机器学习问题。

2 算法

我们提出的算法 *Adam* 的伪代码见算法 1。设 $f(\theta)$ 为带噪声的目标函数: 一个关于参数 θ 可微的随机标量函数。我们希望最小化该函数关于其参数 θ 的期望值, 即 $\mathbb{E}[f(\theta)]$ 。我们用 $f_1(\theta), \dots, f_T(\theta)$ 表示随机函数在后续时间步 $1, \dots, T$ 上的实现值。这种随机性可能来自对数据点随机子样本 (小批量) 的评估, 也可能源于函数固有的噪声。我们用 $g_t = \nabla_{\theta} f_t(\theta)$ 表示梯度, 即 f_t 关于 θ 在时间步 t 处求得的偏导数向量。

该算法更新梯度 (m_t) 和平方梯度 (v_t) 的指数移动平均, 其中超参数 $\beta_1, \beta_2 \in [0, 1]$ 控制这些移动平均的指数衰减率。这些移动平均本身是梯度的 1st 阶矩 (均值) 和 2nd 阶原始矩 (非中心方差) 的估计。然而, 这些移动平均被初始化为 0 (向量), 导致矩估计偏向零, 尤其是在初始时间步, 以及当衰减率较小 (即 β 接近 1) 时更为明显。好消息是, 这种初始化偏置可以很容易地被抵消, 从而得到偏差校正估计 $\hat{m}_t$ 和 $\hat{v}_t$ 。更多细节请参见第 3 节。

请注意, 算法 1 的效率可以通过改变计算顺序来进一步提升, 尽管这会牺牲一定的清晰度, 例如可以将循环中的最后三行替换为以下代码行:

$$\alpha_t = \alpha \cdot \sqrt{1 - \beta_2^t} / (1 - \beta_1^t) \text{ and } \theta_t \leftarrow \theta_{t-1} - \alpha_t \cdot m_t / (\sqrt{v_t} + \epsilon).$$

2.1 Adam 的更新规则

Adam 更新规则的一个重要特性是其对步长的精心选择。假设 $\epsilon = 0$, 则在时间步 t 于参数空间中采取的有效步长为 $\Delta_t = \alpha \cdot \hat{m}_t / \sqrt{\hat{v}_t}$ 。有效步长有两个上界: 在 $(1 - \beta_1) > \sqrt{1 - \beta_2}$ 的情况下为 $|\Delta_t| \leq \alpha \cdot (1 - \beta_1) / \sqrt{1 - \beta_2}$, 否则为 $|\Delta_t| \leq \alpha$

otherwise. The first case only happens in the most severe case of sparsity: when a gradient has been zero at all timesteps except at the current timestep. For less sparse cases, the effective stepsize will be smaller. When $(1 - \beta_1) = \sqrt{1 - \beta_2}$ we have that $|\hat{m}_t/\sqrt{\hat{v}_t}| < 1$ therefore $|\Delta_t| < \alpha$. In more common scenarios, we will have that $\hat{m}_t/\sqrt{\hat{v}_t} \approx \pm 1$ since $|\mathbb{E}[g]/\sqrt{\mathbb{E}[g^2]}| \leq 1$. The effective magnitude of the steps taken in parameter space at each timestep are approximately bounded by the stepsize setting α , i.e., $|\Delta_t| \lesssim \alpha$. This can be understood as establishing a *trust region* around the current parameter value, beyond which the current gradient estimate does not provide sufficient information. This typically makes it relatively easy to know the right scale of α in advance. For many machine learning models, for instance, we often know in advance that good optima are with high probability within some set region in parameter space; it is not uncommon, for example, to have a prior distribution over the parameters. Since α sets (an upper bound of) the magnitude of steps in parameter space, we can often deduce the right order of magnitude of α such that optima can be reached from θ_0 within some number of iterations. With a slight abuse of terminology, we will call the ratio $\hat{m}_t/\sqrt{\hat{v}_t}$ the *signal-to-noise ratio (SNR)*. With a smaller SNR the effective stepsize Δ_t will be closer to zero. This is a desirable property, since a smaller SNR means that there is greater uncertainty about whether the direction of $\hat{m}_t$ corresponds to the direction of the true gradient. For example, the SNR value typically becomes closer to 0 towards an optimum, leading to smaller effective steps in parameter space: a form of automatic annealing. The effective stepsize Δ_t is also invariant to the scale of the gradients; rescaling the gradients g with factor c will scale $\hat{m}_t$ with a factor c and $\hat{v}_t$ with a factor c^2 , which cancel out: $(c \cdot \hat{m}_t)/(\sqrt{c^2 \cdot \hat{v}_t}) = \hat{m}_t/\sqrt{\hat{v}_t}$.

3 INITIALIZATION BIAS CORRECTION

As explained in section 2, Adam utilizes initialization bias correction terms. We will here derive the term for the second moment estimate; the derivation for the first moment estimate is completely analogous. Let g be the gradient of the stochastic objective f , and we wish to estimate its second raw moment (uncentered variance) using an exponential moving average of the squared gradient, with decay rate β_2 . Let $g_1, \dots, g_T$ be the gradients at subsequent timesteps, each a draw from an underlying gradient distribution $g_t \sim p(g_t)$. Let us initialize the exponential moving average as $v_0 = 0$ (a vector of zeros). First note that the update at timestep t of the exponential moving average $v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ (where g_t^2 indicates the elementwise square $g_t \odot g_t$) can be written as a function of the gradients at all previous timesteps:

$$v_t = (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} \cdot g_i^2 \quad (1)$$

We wish to know how $\mathbb{E}[v_t]$, the expected value of the exponential moving average at timestep t , relates to the true second moment $\mathbb{E}[g_t^2]$, so we can correct for the discrepancy between the two. Taking expectations of the left-hand and right-hand sides of eq. (1):

$$\mathbb{E}[v_t] = \mathbb{E} \left[(1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} \cdot g_i^2 \right] \quad (2)$$

$$= \mathbb{E}[g_t^2] \cdot (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} + \zeta \quad (3)$$

$$= \mathbb{E}[g_t^2] \cdot (1 - \beta_2^t) + \zeta \quad (4)$$

where $\zeta = 0$ if the true second moment $\mathbb{E}[g_t^2]$ is stationary; otherwise ζ can be kept small since the exponential decay rate β_1 can (and should) be chosen such that the exponential moving average assigns small weights to gradients too far in the past. What is left is the term $(1 - \beta_2^t)$ which is caused by initializing the running average with zeros. In algorithm 1 we therefore divide by this term to correct the initialization bias.

In case of sparse gradients, for a reliable estimate of the second moment one needs to average over many gradients by choosing a small value of β_2 ; however it is exactly this case of small β_2 where a lack of initialisation bias correction would lead to initial steps that are much larger.

。第一种情况仅在最极端的稀疏性情况下发生：即梯度在当前时间步之前的所有时间步上都为零。对于稀疏程度较低的情况，有效步长会更小。当 $(1 - \beta_1) = \sqrt{1 - \beta_2}$ 时，我们有 $|\hat{m}_t/\sqrt{\hat{v}_t}| < 1$ 因此 $|\Delta_t| < \alpha$ 。在更常见的情形下，由于 $|\mathbb{E}[g]/\sqrt{\mathbb{E}[g^2]}| \leq 1$ ，我们将有 $\hat{m}_t/\sqrt{\hat{v}_t} \approx \pm 1$ 。在每个时间步于参数空间中所采取步长的有效幅度大致受步长设置 α 的约束，即 $|\Delta_t| \lesssim \alpha$ 。这可以理解为在当前参数值附近建立了一个信赖域，超出该范围，当前的梯度估计便无法提供足够的信息。这通常使得人们可以相对容易地预先知道 α 的正确尺度。例如，对于许多机器学习模型，我们通常预先知道好的最优解以很高的概率位于参数空间中的某个特定区域内；例如，对参数设定一个先验分布也并不罕见。由于 α 设定了参数空间中步长的幅度（的上界），我们通常可以推断出 α 的正确数量级，使得最优解能够在若干次迭代内从 θ_0 处到达。稍作术语上的借用，我们将比率 $\hat{m}_t/\sqrt{\hat{v}_t}$ 称为信噪比（SNR）。SNR 越小，有效步长 Δ_t 将越接近零。这是一个理想的性质，因为较小的 SNR 意味着 $\hat{m}_t$ 的方向是否与真实梯度的方向一致存在更大的不确定性。例如，SNR 值通常在接近最优解时趋近于 0，从而导致参数空间中的有效步长变小：这是一种自动退火。有效步长 Δ_t 对梯度的尺度也是不变的；将梯度 g 按因子 c 进行缩放，会使 $\hat{m}_t$ 按因子 c 缩放、 $\hat{v}_t$ 按因子 c^2 缩放，二者相互抵消： $(c \cdot \hat{m}_t)/(\sqrt{c^2 \cdot \hat{v}_t}) = \hat{m}_t/\sqrt{\hat{v}_t}$ 。

3 初始化偏差校正

如第 2 节所述，Adam 使用了初始化偏置校正项。我们将在此推导二阶矩估计的校正项；一阶矩估计的推导过程完全类似。设 g 为随机目标函数 f 的梯度，我们希望利用平方梯度的指数移动平均（衰减率为 β_2 ）来估计其二阶原始矩（非中心方差）。设 $g_1, \dots, g_T$ 为后续各时间步的梯度，每个梯度都是从底层梯度分布 $g_t \sim p(g_t)$ 中抽取的样本。我们将指数移动平均初始化为 $v_0 = 0$ （零向量）。首先注意，时间步 t 上指数移动平均 $v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ 的更新（其中 g_t^2 表示逐元素平方 $g_t \odot g_t$ ）可以写成之前所有时间步梯度的函数：

$$v_t = (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} \cdot g_i^2 \quad (1)$$

我们希望知道 $\mathbb{E}[v_t]$ ，即时间步 t 处指数移动平均的期望值，与真实二阶矩 $\mathbb{E}[g_t^2]$ 之间的关系，以便校正两者之间的差异。对等式 (1) 的左右两边取期望：

$$\mathbb{E}[v_t] = \mathbb{E} \left[(1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} \cdot g_i^2 \right] \quad (2)$$

$$= \mathbb{E}[g_t^2] \cdot (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} + \zeta \quad (3)$$

$$= \mathbb{E}[g_t^2] \cdot (1 - \beta_2^t) + \zeta \quad (4)$$

其中，如果真实二阶矩 $\mathbb{E}[g_i^2]$ 是平稳的，则 $\zeta = 0$ ；否则，由于指数衰减率 β_1 可以（也应该）被选择为使指数移动平均对过于久远的梯度赋予较小权重，因此 ζ 可以保持很小。剩下的项 $(1 - \beta_2^t)$ 是由将运行平均初始化为零引起的。因此，在算法 1 中，我们除以这一项来校正初始化偏置。

在稀疏梯度的情况下，为了对二阶矩进行可靠估计，需要选择较小的 β_2 值来对大量梯度取平均；然而正是在 β_2 较小时，如果缺少初始化偏差校正，会导致初始步骤的步长大得多。

4 CONVERGENCE ANALYSIS

We analyze the convergence of Adam using the online learning framework proposed in (Zinkevich, 2003). Given an arbitrary, unknown sequence of convex cost functions $f_1(\theta), f_2(\theta), \dots, f_T(\theta)$. At each time t , our goal is to predict the parameter θ_t and evaluate it on a previously unknown cost function f_t . Since the nature of the sequence is unknown in advance, we evaluate our algorithm using the regret, that is the sum of all the previous difference between the online prediction $f_t(\theta_t)$ and the best fixed point parameter $f_t(\theta^*)$ from a feasible set $\mathcal{X}$ for all the previous steps. Concretely, the regret is defined as:

$$R(T) = \sum_{t=1}^T [f_t(\theta_t) - f_t(\theta^*)] \quad (5)$$

where $\theta^* = \arg \min_{\theta \in \mathcal{X}} \sum_{t=1}^T f_t(\theta)$. We show Adam has $O(\sqrt{T})$ regret bound and a proof is given in the appendix. Our result is comparable to the best known bound for this general convex online learning problem. We also use some definitions simplify our notation, where $g_t \triangleq \nabla f_t(\theta_t)$ and $g_{t,i}$ as the i^{th} element. We define $g_{1:t,i} \in \mathbb{R}^t$ as a vector that contains the i^{th} dimension of the gradients over all iterations till t , $g_{1:t,i} = [g_{1,i}, g_{2,i}, \dots, g_{t,i}]$. Also, we define $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. Our following theorem holds when the learning rate α_t is decaying at a rate of $t^{-\frac{1}{2}}$ and first moment running average coefficient $\beta_{1,t}$ decay exponentially with λ , that is typically close to 1, e.g. $1 - 10^{-8}$.

Theorem 4.1. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in \mathbb{R}^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, \dots, T\}$, and $\beta_1, \beta_2 \in [0, 1)$ satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$. Let $\alpha_t = \frac{\alpha}{\sqrt{t}}$ and $\beta_{1,t} = \beta_1 \lambda^{t-1}$, $\lambda \in (0, 1)$. Adam achieves the following guarantee, for all $T \geq 1$.*

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T \hat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \sum_{i=1}^d \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Our Theorem 4.1 implies when the data features are sparse and bounded gradients, the summation term can be much smaller than its upper bound $\sum_{i=1}^d \|g_{1:T,i}\|_2 << dG_\infty\sqrt{T}$ and $\sum_{i=1}^d \sqrt{T \hat{v}_{T,i}} << dG_\infty\sqrt{T}$, in particular if the class of function and data features are in the form of section 1.2 in (Duchi et al., 2011). Their results for the expected value $\mathbb{E}[\sum_{i=1}^d \|g_{1:T,i}\|_2]$ also apply to Adam. In particular, the adaptive method, such as Adam and Adagrad, can achieve $O(\log d\sqrt{T})$, an improvement over $O(\sqrt{dT})$ for the non-adaptive method. Decaying $\beta_{1,t}$ towards zero is important in our theoretical analysis and also matches previous empirical findings, e.g. (Sutskever et al., 2013) suggests reducing the momentum coefficient in the end of training can improve convergence.

Finally, we can show the average regret of Adam converges,

Corollary 4.2. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in \mathbb{R}^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, \dots, T\}$. Adam achieves the following guarantee, for all $T \geq 1$.*

$$\frac{R(T)}{T} = O\left(\frac{1}{\sqrt{T}}\right)$$

This result can be obtained by using Theorem 4.1 and $\sum_{i=1}^d \|g_{1:T,i}\|_2 \leq dG_\infty\sqrt{T}$. Thus, $\lim_{T \rightarrow \infty} \frac{R(T)}{T} = 0$.

5 RELATED WORK

Optimization methods bearing a direct relation to Adam are RMSProp (Tieleman & Hinton, 2012; Graves, 2013) and AdaGrad (Duchi et al., 2011); these relationships are discussed below. Other stochastic optimization methods include vSGD (Schaul et al., 2012), AdaDelta (Zeiler, 2012) and the natural Newton method from Roux & Fitzgibbon (2010), all setting stepsizes by estimating curvature

4 收敛性分析

我们采用 (Zinkevich, 2003) 提出的在线学习框架来分析 Adam 的收敛性。给定一个任意的、未知的凸成本函数序列 $f_1(\theta)$ 、 $f_2(\theta)$ 、 $\dots$ 、 $f_T(\theta)$ ，在每个时刻 t ，我们的目标是预测参数 θ_t ，并在之前未知的成本函数 f_t 上对其进行评估。由于序列的性质事先未知，我们使用遗憾值来评估算法，即在所有先前步骤中，在线预测 $f_t(\theta_t)$ 与来自可行集 $\mathcal{X}$ 的最佳固定点参数 $f_t(\theta^*)$ 之间所有差值的总和。具体而言，遗憾值定义为：

$$R(T) = \sum_{t=1}^T [f_t(\theta_t) - f_t(\theta^*)] \quad (5)$$

其中 $\theta^* = \arg \min_{\theta \in \mathcal{X}} \sum_{t=1}^T f_t(\theta)$ 。我们证明了 Adam 具有 $O(\sqrt{T})$ 遗憾界，相关证明见附录。我们的结果可与该一般凸在线学习问题目前已知的最佳界相媲美。我们还使用一些定义来简化符号，其中 $g_t \triangleq \nabla f_t(\theta_t)$ 和 $g_{t,i}$ 表示第 i^{th} 个元素。我们定义 $g_{1:t,i} \in \mathbb{R}^t$ 为一个向量，它包含截至 t 所有迭代中梯度的第 i^{th} 维，即 $g_{1:t,i} = [g_{1,i}, g_{2,i}, \dots, g_{t,i}]$ 。此外，我们定义 $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$ 。当学习率 α_t 以 $t^{-\frac{1}{2}}$ 的速率衰减，且一阶矩滑动平均系数 $\beta_{1,t}$ 随 λ 指数衰减时（ λ 通常接近 1，例如 $1 - 10^{-8}$ ），以下定理成立。

定理 4.1 f_t θ_t $\|\nabla f_t \theta\|_2 \leq G$ $\|\nabla f_t \theta\|_\infty \leq \dots$ 假设函数具有有界梯度， $(\cdot), (\cdot)$ 对于所有成立，并且 Adam 生成的任意之间的距离有界， $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ 对于任意 $m, n \in \{1, \dots, T\}$ ，以及 $\beta_1, \beta_2 \in [0, 1)$ 满足 $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$ 。令 $\alpha_t = \frac{\alpha}{\sqrt{t}}$ 且 $\beta_{1,t} = \beta_1 \lambda^{t-1}, \lambda \in (0, 1)$ 。Adam 对所有 $T \geq 1$ 实现如下保证。

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T \hat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \sum_{i=1}^d \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

我们的定理 4.1 表明，当数据特征稀疏且梯度有界时，求和项可以远小于其上界 $\sum_{i=1}^d \|g_{1:T,i}\|_2 << dG_\infty \sqrt{T}$ 和 $\sum_{i=1}^d \sqrt{T \hat{v}_{T,i}} << dG_\infty \sqrt{T}$ ；特别地，如果函数类别和数据特征符合 (Duchi et al., 2011) 第 1.2 节中的形式。他们关于期望值 $\mathbb{E}[\sum_{i=1}^d \|g_{1:T,i}\|_2]$ 的结果同样适用于 Adam。尤其是，Adam 和 AdaGrad 等自适应方法可以达到 $O(\log d\sqrt{T})$ ，相较于非自适应方法的 $O(\sqrt{dT})$ 有所改进。将 $\beta_{1,t}$ 向零衰减在我们的理论分析中很重要，也与以往的实证发现一致；例如 (Sutskever et al., 2013) 建议在训练末期降低动量系数可以改进收敛。

最后，我们可以证明 Adam 的平均遗憾值收敛，

推论 4.2 f_t θ_t $\|\nabla f_t \theta\|_2 \leq G$ $\|\nabla f_t \theta\|_\infty \leq \dots$ 假设该函数具有有界梯度， $(\cdot), (\cdot)$ 对所有成立，且 Adam 生成的任意之间的距离有界， $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ 对任意 $m, n \in \{1, \dots, T\}$ 成立。Adam 对所有 $T \geq 1$ 达到以下保证。

$$\frac{R(T)}{T} = O\left(\frac{1}{\sqrt{T}}\right)$$

该结果可以通过使用定理 4.1 和 $\sum_{i=1}^d \|g_{1:T,i}\|_2 \leq dG_\infty \sqrt{T}$ 得到。因此， $\lim_{T \rightarrow \infty} \frac{R(T)}{T} = 0$ 。

5 相关工作

与 Adam 有直接关系的优化方法包括 RMSProp (Tieleman 与 Hinton, 2012; Graves, 2013) 和 AdaGrad (Duchi 等人, 2011)；这些关系将在下文讨论。其他随机优化方法包括 vSGD (Schaul 等人, 2012)、AdaDelta (Zeiler, 2012) 以及 Roux 与 Fitzgibbon (2010) 提出的自然牛顿法，这些方法都是通过估计曲率来设置步长，

from first-order information. The Sum-of-Functions Optimizer (SFO) (Sohl-Dickstein et al., 2014) is a quasi-Newton method based on minibatches, but (unlike Adam) has memory requirements linear in the number of minibatch partitions of a dataset, which is often infeasible on memory-constrained systems such as a GPU. Like natural gradient descent (NGD) (Amari, 1998), Adam employs a preconditioner that adapts to the geometry of the data, since $\hat{v}_t$ is an approximation to the diagonal of the Fisher information matrix (Pascanu & Bengio, 2013); however, Adam’s preconditioner (like AdaGrad’s) is more conservative in its adaption than vanilla NGD by preconditioning with the square root of the inverse of the diagonal Fisher information matrix approximation.

RMSPProp: An optimization method closely related to Adam is RMSPProp (Tieleman & Hinton, 2012). A version with momentum has sometimes been used (Graves, 2013). There are a few important differences between RMSPProp with momentum and Adam: RMSPProp with momentum generates its parameter updates using a momentum on the rescaled gradient, whereas Adam updates are directly estimated using a running average of first and second moment of the gradient. RMSPProp also lacks a bias-correction term; this matters most in case of a value of β_2 close to 1 (required in case of sparse gradients), since in that case not correcting the bias leads to very large stepsizes and often divergence, as we also empirically demonstrate in section 6.4.

AdaGrad: An algorithm that works well for sparse gradients is AdaGrad (Duchi et al., 2011). Its basic version updates parameters as $\theta_{t+1} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^t g_i^2}$. Note that if we choose β_2 to be infinitesimally close to 1 from below, then $\lim_{\beta_2 \rightarrow 1} \hat{v}_t = t^{-1} \cdot \sum_{i=1}^t g_i^2$. AdaGrad corresponds to a version of Adam with $\beta_1 = 0$, infinitesimal $(1 - \beta_2)$ and a replacement of α by an annealed version $\alpha_t = \alpha \cdot t^{-1/2}$, namely $\theta_t - \alpha \cdot t^{-1/2} \cdot \hat{m}_t / \sqrt{\lim_{\beta_2 \rightarrow 1} \hat{v}_t} = \theta_t - \alpha \cdot t^{-1/2} \cdot g_t / \sqrt{t^{-1} \cdot \sum_{i=1}^t g_i^2} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^t g_i^2}$. Note that this direct correspondence between Adam and Adagrad does not hold when removing the bias-correction terms; without bias correction, like in RMSPProp, a β_2 infinitesimally close to 1 would lead to infinitely large bias, and infinitely large parameter updates.

6 EXPERIMENTS

To empirically evaluate the proposed method, we investigated different popular machine learning models, including logistic regression, multilayer fully connected neural networks and deep convolutional neural networks. Using large models and datasets, we demonstrate Adam can efficiently solve practical deep learning problems.

We use the same parameter initialization when comparing different optimization algorithms. The hyper-parameters, such as learning rate and momentum, are searched over a dense grid and the results are reported using the best hyper-parameter setting.

6.1 EXPERIMENT: LOGISTIC REGRESSION

We evaluate our proposed method on L2-regularized multi-class logistic regression using the MNIST dataset. Logistic regression has a well-studied convex objective, making it suitable for comparison of different optimizers without worrying about local minimum issues. The stepsize α in our logistic regression experiments is adjusted by $1/\sqrt{t}$ decay, namely $\alpha_t = \frac{\alpha}{\sqrt{t}}$ that matches with our theoretical prediction from section 4. The logistic regression classifies the class label directly on the 784 dimension image vectors. We compare Adam to accelerated SGD with Nesterov momentum and Adagrad using minibatch size of 128. According to Figure 1, we found that the Adam yields similar convergence as SGD with momentum and both converge faster than Adagrad.

As discussed in (Duchi et al., 2011), Adagrad can efficiently deal with sparse features and gradients as one of its main theoretical results whereas SGD is low at learning rare features. Adam with $1/\sqrt{t}$ decay on its stepsize should theoretically match the performance of Adagrad. We examine the sparse feature problem using IMDB movie review dataset from (Maas et al., 2011). We pre-process the IMDB movie reviews into bag-of-words (BoW) feature vectors including the first 10,000 most frequent words. The 10,000 dimension BoW feature vector for each review is highly sparse. As suggested in (Wang & Manning, 2013), 50% dropout noise can be applied to the BoW features during

其依据是一阶信息。函数和优化器 (SFO) (Sohl-Dickstein 等人, 2014) 是一种基于小批量的拟牛顿法, 但 (与 Adam 不同) 其内存需求与数据集小批量划分的数量成线性关系, 这在 GPU 等内存受限的系统上往往不可行。与自然梯度下降 (NGD) (Amari, 1998) 类似, Adam 使用了一种能适应数据几何结构的预条件子, 因为 $\hat{v}_t$ 是费希尔信息矩阵对角线的近似 (Pascanu 与 Bengio, 2013); 然而, Adam 的预条件子 (与 AdaGrad 的一样) 通过使用对角费希尔信息矩阵近似之逆的平方根进行预条件处理, 使其适应性比普通 NGD 更为保守。

RMSProp: 与 Adam 密切相关的一种优化方法是 RMSProp (Tieleman & Hinton, 2012)。其带动量版本有时也被使用 (Graves, 2013)。带动量的 RMSProp 与 Adam 之间存在若干重要差异: 带动量的 RMSProp 通过对重缩放梯度施加动量来生成参数更新, 而 Adam 的更新则是直接使用梯度的一阶和二阶矩的运行平均来估计。RMSProp 还缺少偏差校正项; 这在 β_2 接近 1 的情况下影响最大 (稀疏梯度情况下需要如此), 因为此时若不校正偏差, 会导致非常大的步长并常常发散, 我们也在 6.4 节中通过实验证明了这一点。

AdaGrad: AdaGrad (Duchi et al., 2011) 是一种对稀疏梯度表现良好的算法。其基本版本按 $\theta_{t+1} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^t g_i^2}$ 更新参数。注意, 如果我们选择 β_2 从下方无限趋近于 1, 则 $\lim_{\beta_2 \rightarrow 1} \hat{v}_t = t^{-1} \cdot \sum_{i=1}^t g_i^2$ 。AdaGrad 对应于 Adam 的一个版本: $\beta_1 = 0$ 、无穷小的 $(1 - \beta_2)$, 并将 α 替换为退火版本 $\alpha_t = \alpha \cdot t^{-1/2}$, 即 $\theta_t - \alpha \cdot t^{-1/2} \cdot \hat{m}_t / \sqrt{\lim_{\beta_2 \rightarrow 1} \hat{v}_t} = \theta_t - \alpha \cdot t^{-1/2} \cdot g_t / \sqrt{t^{-1} \cdot \sum_{i=1}^t g_i^2} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^t g_i^2}$ 。注意, 当移除偏置校正项时, Adam 与 AdaGrad 之间的这种直接对应关系不再成立; 与 RMSProp 一样, 若没有偏置校正, β_2 无限趋近于 1 会导致无限大的偏差和无限大的参数更新。

6 实验

为了实证评估所提出的方法, 我们研究了多种流行的机器学习模型, 包括 Logistic 回归、多层全连接神经网络以及深度卷积神经网络。通过使用大型模型和数据集, 我们证明 Adam 能够高效地解决实际的深度学习问题。

在比较不同优化算法时, 我们使用相同的参数初始化。学习率、动量等超参数均在密集网格上搜索, 并报告最佳超参数设置下的结果。

6.1 实验: LOGISTIC 回归

我们使用 MNIST 数据集, 在 L2 正则化的多分类 Logistic 回归上评估所提出的方法。Logistic 回归具有已被充分研究的凸目标, 因此适合用于比较不同优化器, 而无需担心局部最小问题。在我们的 Logistic 回归实验中, 步长 α 通过 $1/\sqrt{t}$ 衰减进行调整, 即 $\alpha_t = \frac{\alpha}{\sqrt{t}}$, 这与第 4 节中的理论预测一致。该 Logistic 回归直接在 784 维图像向量上对类别标签进行分类。我们将 Adam 与采用 Nesterov 动量的加速 SGD 以及 Adagrad 进行了比较, 小批量大小为 128。根据图 1, 我们发现 Adam 的收敛速度与带动量的 SGD 相近, 且两者均比 Adagrad 收敛得更快。

如 (Duchi et al., 2011) 所述, AdaGrad 能够有效处理稀疏特征和稀疏梯度, 这是其主要的理论成果之一, 而 SGD 在学习稀有特征方面表现较差。理论上, 带有 $1/\sqrt{t}$ 步长衰减的 Adam 应与 AdaGrad 的性能相当。我们使用来自 (Maas et al., 2011) 的 IMDB 电影评论数据集来考察稀疏特征问题。我们将 IMDB 电影评论预处理为词袋 (BoW) 特征向量, 其中包含出现频率最高的前 10,000 个单词。每条评论的 10,000 维词袋特征向量是高度稀疏的。如 (Wang & Manning, 2013) 所建议的, 可以在训练过程中对词袋特征施加 50% 的 dropout 噪声,

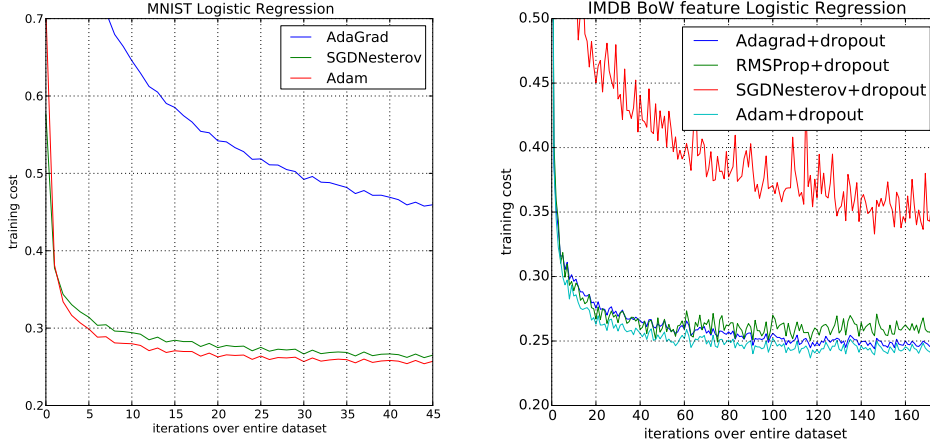


Figure 1: Logistic regression training negative log likelihood on MNIST images and IMDB movie reviews with 10,000 bag-of-words (BoW) feature vectors.

training to prevent over-fitting. In figure 1, Adagrad outperforms SGD with Nesterov momentum by a large margin both with and without dropout noise. Adam converges as fast as Adagrad. The empirical performance of Adam is consistent with our theoretical findings in sections 2 and 4. Similar to Adagrad, Adam can take advantage of sparse features and obtain faster convergence rate than normal SGD with momentum.

6.2 EXPERIMENT: MULTI-LAYER NEURAL NETWORKS

Multi-layer neural network are powerful models with non-convex objective functions. Although our convergence analysis does not apply to non-convex problems, we empirically found that Adam often outperforms other methods in such cases. In our experiments, we made model choices that are consistent with previous publications in the area; a neural network model with two fully connected hidden layers with 1000 hidden units each and ReLU activation are used for this experiment with minibatch size of 128.

First, we study different optimizers using the standard deterministic cross-entropy objective function with L_2 weight decay on the parameters to prevent over-fitting. The sum-of-functions (SFO) method (Sohl-Dickstein et al., 2014) is a recently proposed quasi-Newton method that works with minibatches of data and has shown good performance on optimization of multi-layer neural networks. We used their implementation and compared with Adam to train such models. Figure 2 shows that Adam makes faster progress in terms of both the number of iterations and wall-clock time. Due to the cost of updating curvature information, SFO is 5-10x slower per iteration compared to Adam, and has a memory requirement that is linear in the number minibatches.

Stochastic regularization methods, such as dropout, are an effective way to prevent over-fitting and often used in practice due to their simplicity. SFO assumes deterministic subfunctions, and indeed failed to converge on cost functions with stochastic regularization. We compare the effectiveness of Adam to other stochastic first order methods on multi-layer neural networks trained with dropout noise. Figure 2 shows our results; Adam shows better convergence than other methods.

6.3 EXPERIMENT: CONVOLUTIONAL NEURAL NETWORKS

Convolutional neural networks (CNNs) with several layers of convolution, pooling and non-linear units have shown considerable success in computer vision tasks. Unlike most fully connected neural nets, weight sharing in CNNs results in vastly different gradients in different layers. A smaller learning rate for the convolution layers is often used in practice when applying SGD. We show the effectiveness of Adam in deep CNNs. Our CNN architecture has three alternating stages of 5x5 convolution filters and 3x3 max pooling with stride of 2 that are followed by a fully connected layer of 1000 rectified linear hidden units (ReLU's). The input image are pre-processed by whitening, and

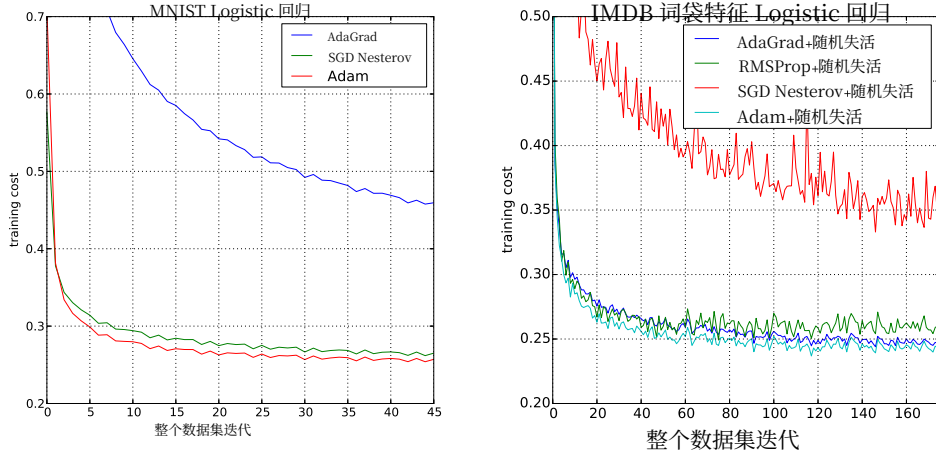


图 1: 在 MNIST 图像和包含 10,000 个词袋 (BoW) 特征向量的 IMDB 电影评论上训练 Logistic 回归的负对数似然。

以防止过拟合。在图 1 中，无论是否使用 dropout 噪声，AdaGrad 都以很大优势优于使用 Nesterov 动量的 SGD。Adam 的收敛速度与 AdaGrad 一样快。Adam 的实验表现与我们在第 2 节和第 4 节中的理论发现一致。与 AdaGrad 类似，Adam 能够利用稀疏特征，并获得比带动量的普通 SGD 更快的收敛速率。

6.2 实验：多层神经网络

多层神经网络是具有非凸目标函数的强大模型。尽管我们的收敛性分析并不适用于非凸问题，但我们从经验上发现，Adam 在这类情况下往往优于其他方法。在我们的实验中，模型选择与该领域先前的公开成果保持一致；本实验采用一个包含两个全连接隐藏层、每层 1000 个隐藏单元并使用 ReLU 激活函数的神经网络模型，小批量大小为 128。

首先，我们使用标准的确定性交叉熵目标函数研究不同的优化器，并对参数施加 L_2 权重衰减以防止过拟合。函数和 (SFO) 方法 (Sohl-Dickstein 等人, 2014) 是最近提出的一种拟牛顿法，可处理小批量数据，并已在多层神经网络的优化中表现出良好性能。我们使用了他们的实现，并将其与 Adam 进行比较来训练此类模型。图 2 显示，无论从迭代次数还是实际耗时来看，Adam 的收敛速度都更快。由于更新曲率信息的代价，SFO 每次迭代的耗时比 Adam 慢 5-10 倍，且其内存需求随小批量数量线性增长。

随机正则化方法（如随机失活）是防止过拟合的有效手段，因其简单性而在实践中被广泛使用。SFO 假定子函数是确定性的，因此在带有随机正则化的成本函数上确实无法收敛。我们在使用 dropout 噪声训练的多层神经网络上，比较了 Adam 与其他随机一阶方法的效果。图 2 展示了我们的结果；Adam 表现出比其他方法更好的收敛性。

6.3 实验：卷积神经网络

卷积神经网络 (CNN) 由若干卷积层、池化层和非线性单元组成，已在计算机视觉任务中取得了相当大的成功。与大多数全连接神经网络不同，CNN 中的权共享导致不同层的梯度差异巨大。在实践中应用 SGD 时，通常会对卷积层使用较小的学习率。我们展示了 Adam 在深度 CNN 中的有效性。我们的 CNN 架构包含三个交替的阶段，每个阶段由 5×5 卷积滤波器和步幅为 2 的 3×3 最大池化组成，之后是一个包含 1000 个修正线性隐藏单元 (ReLU) 的全连接层。输入图像经过白化预处理，并且

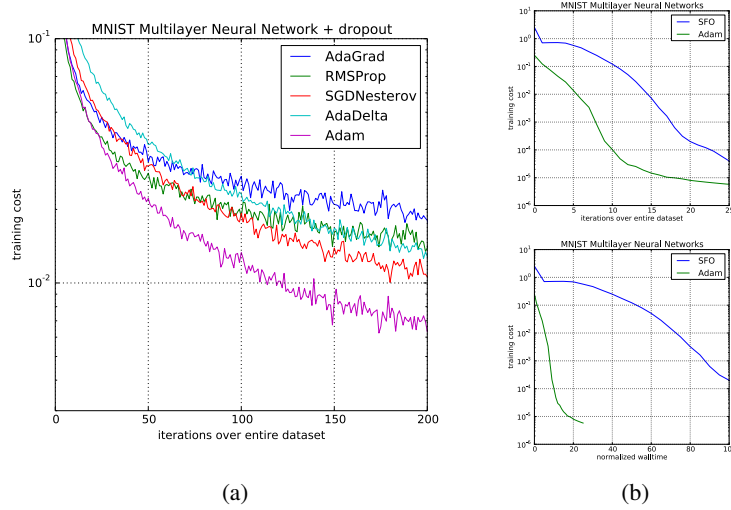


Figure 2: Training of multilayer neural networks on MNIST images. (a) Neural networks using dropout stochastic regularization. (b) Neural networks with deterministic cost function. We compare with the sum-of-functions (SFO) optimizer (Sohl-Dickstein et al., 2014)

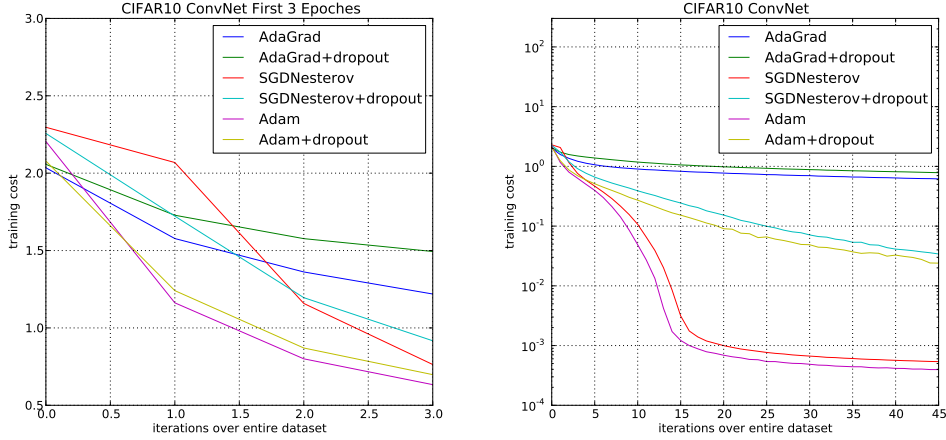


Figure 3: Convolutional neural networks training cost. (left) Training cost for the first three epochs. (right) Training cost over 45 epochs. CIFAR-10 with c64-c64-c128-1000 architecture.

dropout noise is applied to the input layer and fully connected layer. The minibatch size is also set to 128 similar to previous experiments.

Interestingly, although both Adam and Adagrad make rapid progress lowering the cost in the initial stage of the training, shown in Figure 3 (left), Adam and SGD eventually converge considerably faster than Adagrad for CNNs shown in Figure 3 (right). We notice the second moment estimate $\hat{v}_t$ vanishes to zeros after a few epochs and is dominated by the ϵ in algorithm 1. The second moment estimate is therefore a poor approximation to the geometry of the cost function in CNNs comparing to fully connected network from Section 6.2. Whereas, reducing the minibatch variance through the first moment is more important in CNNs and contributes to the speed-up. As a result, Adagrad converges much slower than others in this particular experiment. Though Adam shows marginal improvement over SGD with momentum, it adapts learning rate scale for different layers instead of hand picking manually as in SGD.

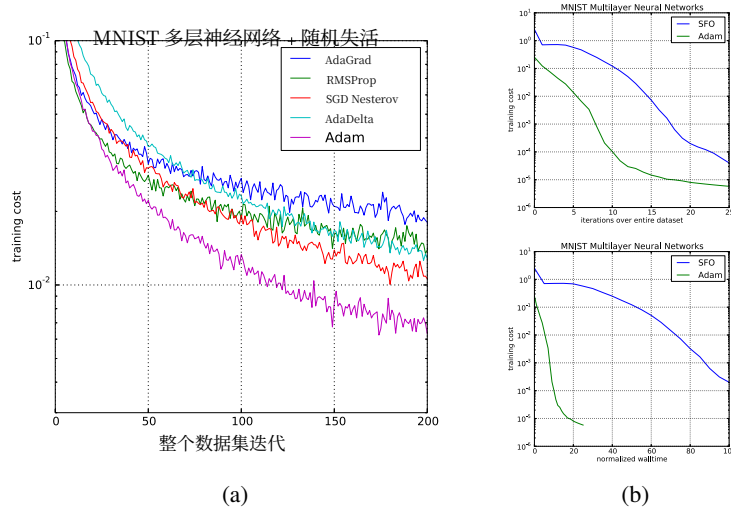


图 2: 在 MNIST 图像上训练多层神经网络。(a) 使用 dropout 随机正则化的神经网络。(b) 采用确定性成本函数的神经网络。我们与函数和 (SFO) 优化器 (Sohl-Dickstein 等人, 2014) 进行了比较

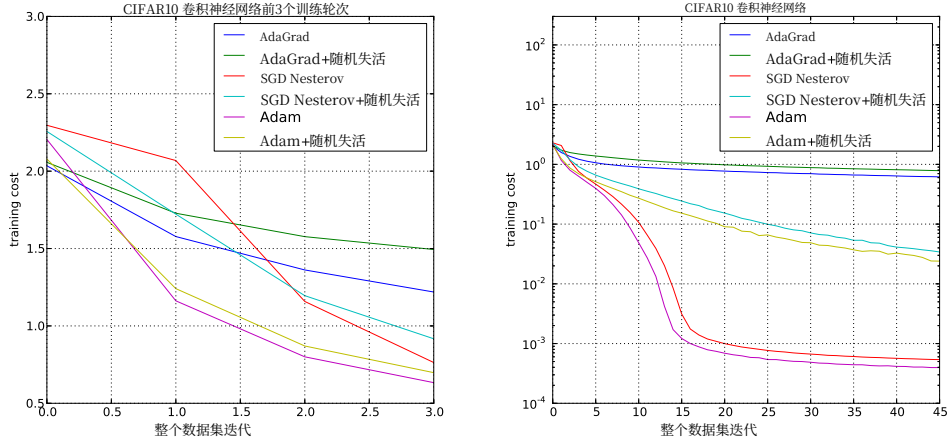


图 3: 卷积神经网络训练成本。(左) 前三个轮次的训练成本。(右) 45 个轮次内的训练成本。采用 c64-c64-c128-1000 架构的 CIFAR-10。

dropout 噪声被应用于输入层和全连接层。与之前的实验类似，小批量大小也设置为 128。

有趣的是，尽管 Adam 和 Adagrad 在训练初期都迅速降低了成本，如图 3（左）所示，但对于 CNN，Adam 和 SGD 最终的收敛速度明显快于 Adagrad，如图 3（右）所示。我们注意到，二阶矩估计 $\hat{v}_t$ 在几个轮次后变为零，并且在算法 1 中被 ϵ 主导。因此，与第 6.2 节中的全连接网络相比，CNN 中的二阶矩估计对成本函数几何结构的近似效果较差。而通过一阶矩降低小批量方差在 CNN 中更为重要，并有助于加速。结果，在这个特定实验中，Adagrad 的收敛速度远慢于其他方法。尽管 Adam 相比带动量的 SGD 仅有边际改进，但它会为不同层自适应调整学习率尺度，而不是像 SGD 那样手动选取。

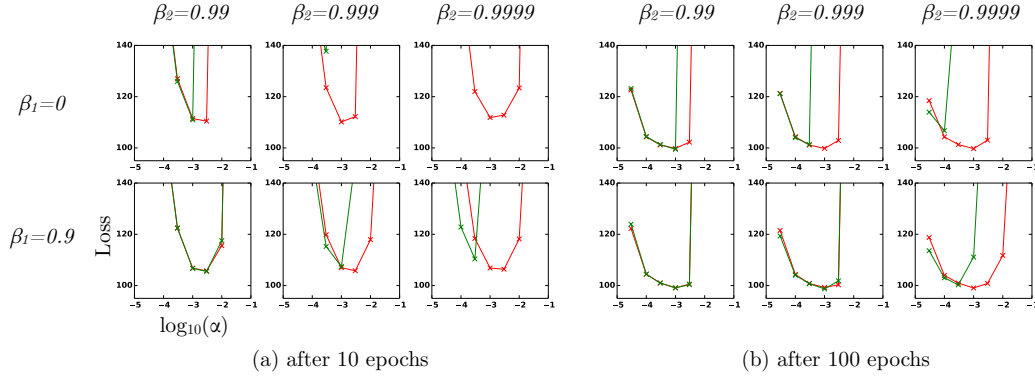


Figure 4: Effect of bias-correction terms (red line) versus no bias correction terms (green line) after 10 epochs (left) and 100 epochs (right) on the loss (y-axes) when learning a Variational Auto-Encoder (VAE) (Kingma & Welling, 2013), for different settings of stepsize α (x-axes) and hyper-parameters β_1 and β_2 .

6.4 EXPERIMENT: BIAS-CORRECTION TERM

We also empirically evaluate the effect of the bias correction terms explained in sections 2 and 3. Discussed in section 5, removal of the bias correction terms results in a version of RMSProp (Tieleman & Hinton, 2012) with momentum. We vary the β_1 and β_2 when training a variational auto-encoder (VAE) with the same architecture as in (Kingma & Welling, 2013) with a single hidden layer with 500 hidden units with softplus nonlinearities and a 50-dimensional spherical Gaussian latent variable. We iterated over a broad range of hyper-parameter choices, i.e. $\beta_1 \in [0, 0.9]$ and $\beta_2 \in [0.99, 0.999, 0.9999]$, and $\log_{10}(\alpha) \in [-5, \dots, -1]$. Values of β_2 close to 1, required for robustness to sparse gradients, results in larger initialization bias; therefore we expect the bias correction term is important in such cases of slow decay, preventing an adverse effect on optimization.

In Figure 4, values β_2 close to 1 indeed lead to instabilities in training when no bias correction term was present, especially at first few epochs of the training. The best results were achieved with small values of $(1 - \beta_2)$ and bias correction; this was more apparent towards the end of optimization when gradients tends to become sparser as hidden units specialize to specific patterns. In summary, Adam performed equal or better than RMSProp, regardless of hyper-parameter setting.

7 EXTENSIONS

7.1 ADAMAX

In Adam, the update rule for individual weights is to scale their gradients inversely proportional to a (scaled) L^2 norm of their individual current and past gradients. We can generalize the L^2 norm based update rule to a L^p norm based update rule. Such variants become numerically unstable for large p . However, in the special case where we let $p \rightarrow \infty$, a surprisingly simple and stable algorithm emerges; see algorithm 2. We'll now derive the algorithm. Let, in case of the L^p norm, the stepsize at time t be inversely proportional to $v_t^{1/p}$, where:

$$v_t = \beta_2^p v_{t-1} + (1 - \beta_2^p) |g_t|^p \quad (6)$$

$$= (1 - \beta_2^p) \sum_{i=1}^t \beta_2^{p(t-i)} \cdot |g_i|^p \quad (7)$$

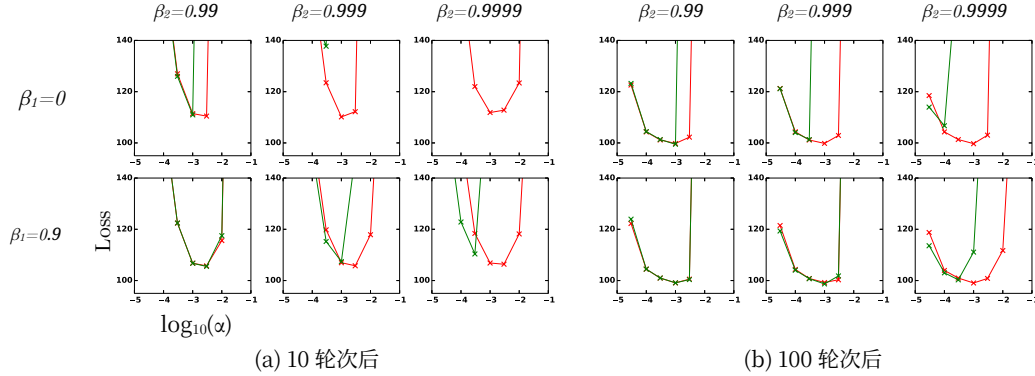


图 4: 在学习变分自编码器 (VAE) (Kingma & Welling, 2013) 时, 偏置校正项 (红线) 与无偏置校正项 (绿线) 在 10 轮次 (左) 和 100 轮次 (右) 后对损失 (y 轴) 的影响, 其中包含不同的步长 α (x 轴) 以及超参数 β_1 和 β_2 设置。

6.4 实验: 偏置校正项

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我们还通过实验评估了第 2 节和第 3 节中解释的偏置校正项的影响。如第 5 节所述, 移除偏置校正项会得到一个带动量的 RMSProp (Tieleman & Hinton, 2012) 版本。我们在训练变分自编码器 (VAE) 时改变 β_1 和 β_2 , 该 VAE 采用与 (Kingma & Welling, 2013) 相同的架构: 单个隐藏层包含 500 个带有 softplus 非线性的隐藏单元, 以及一个 50 维球面高斯隐变量。我们遍历了范围很广的超参数选择, 即 $\beta_1 \in [0, 0.9]$ 和 $\beta_2 \in [0.99, 0.999, 0.9999]$, 以及 $\log_{10}(\alpha) \in [-5, \dots, -1]$ 。接近 1 的 β_2 值是实现对稀疏梯度鲁棒性所必需的, 但会带来更大的初始化偏置; 因此我们预计, 在这种衰减缓慢的情况下, 偏置校正项非常重要, 可以防止其对优化产生不利影响。

在图 4 中, 当不存在偏置校正项时, 接近 1 的 β_2 值确实会导致训练不稳定, 尤其是在训练的最初几个轮次。使用较小的 $(1 - \beta_2)$ 值和偏置校正可以获得最佳结果; 这一点在优化的后期更为明显, 此时随着隐藏单元专门化于特定模式, 梯度趋于变得更加稀疏。总之, 无论超参数设置如何, Adam 的表现都与 RMSProp 相当或更好。

7 扩展

7.1 AdaMax

在 Adam 中, 单个权重的更新规则是将其梯度按与各自当前及过去梯度的 (缩放) L^2 范数成反比的方式进行缩放。我们可以将基于 L^2 范数的更新规则推广到基于 L^p 范数的更新规则。此类变体在 p 较大时会数值不稳定。然而, 在特殊情况下令 $p \rightarrow \infty$, 会出现一个出人意料简单且稳定的算法; 见算法 2。现在推导该算法。令在 L^p 范数情形下, 时间 t 的步长与 $v_t^{1/p}$ 成反比, 其中:

$$v_t = \beta_2^p v_{t-1} + (1 - \beta_2^p) |g_t|^p \quad (6)$$

$$= (1 - \beta_2^p) \sum_{i=1}^t \beta_2^{p(t-i)} \cdot |g_i|^p \quad (7)$$

Algorithm 2: *AdaMax*, a variant of Adam based on the infinity norm. See section 7.1 for details. Good default settings for the tested machine learning problems are $\alpha = 0.002$, $\beta_1 = 0.9$ and $\beta_2 = 0.999$. With β_1^t we denote β_1 to the power t . Here, $(\alpha/(1 - \beta_1^t))$ is the learning rate with the bias-correction term for the first moment. All operations on vectors are element-wise.

Require: α : Stepsize

Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates

Require: $f(\theta)$: Stochastic objective function with parameters θ

Require: θ_0 : Initial parameter vector

$m_0 \leftarrow 0$ (Initialize 1st moment vector)

$u_0 \leftarrow 0$ (Initialize the exponentially weighted infinity norm)

$t \leftarrow 0$ (Initialize timestep)

while θ_t not converged **do**

$t \leftarrow t + 1$

$g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)

$m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)

$u_t \leftarrow \max(\beta_2 \cdot u_{t-1}, |g_t|)$ (Update the exponentially weighted infinity norm)

$\theta_t \leftarrow \theta_{t-1} - (\alpha/(1 - \beta_1^t)) \cdot m_t/u_t$ (Update parameters)

end while

return θ_t (Resulting parameters)

Note that the decay term is here equivalently parameterised as β_2^p instead of β_2 . Now let $p \rightarrow \infty$, and define $u_t = \lim_{p \rightarrow \infty} (v_t)^{1/p}$, then:

$$u_t = \lim_{p \rightarrow \infty} (v_t)^{1/p} = \lim_{p \rightarrow \infty} \left((1 - \beta_2^p) \sum_{i=1}^t \beta_2^{p(t-i)} \cdot |g_i|^p \right)^{1/p} \quad (8)$$

$$= \lim_{p \rightarrow \infty} (1 - \beta_2^p)^{1/p} \left(\sum_{i=1}^t \beta_2^{p(t-i)} \cdot |g_i|^p \right)^{1/p} \quad (9)$$

$$= \lim_{p \rightarrow \infty} \left(\sum_{i=1}^t \left(\beta_2^{(t-i)} \cdot |g_i| \right)^p \right)^{1/p} \quad (10)$$

$$= \max(\beta_2^{t-1} |g_1|, \beta_2^{t-2} |g_2|, \dots, \beta_2 |g_{t-1}|, |g_t|) \quad (11)$$

Which corresponds to the remarkably simple recursive formula:

$$u_t = \max(\beta_2 \cdot u_{t-1}, |g_t|) \quad (12)$$

with initial value $u_0 = 0$. Note that, conveniently enough, we don't need to correct for initialization bias in this case. Also note that the magnitude of parameter updates has a simpler bound with AdaMax than Adam, namely: $|\Delta_t| \leq \alpha$.

7.2 TEMPORAL AVERAGING

Since the last iterate is noisy due to stochastic approximation, better generalization performance is often achieved by averaging. Previously in Moulines & Bach (2011), Polyak-Ruppert averaging (Polyak & Juditsky, 1992; Ruppert, 1988) has been shown to improve the convergence of standard SGD, where $\bar{\theta}_t = \frac{1}{t} \sum_{k=1}^t \theta_k$. Alternatively, an exponential moving average over the parameters can be used, giving higher weight to more recent parameter values. This can be trivially implemented by adding one line to the inner loop of algorithms 1 and 2: $\bar{\theta}_t \leftarrow \beta_2 \cdot \bar{\theta}_{t-1} + (1 - \beta_2) \theta_t$, with $\bar{\theta}_0 = 0$. Initialization bias can again be corrected by the estimator $\hat{\theta}_t = \bar{\theta}_t / (1 - \beta_2^t)$.

8 CONCLUSION

We have introduced a simple and computationally efficient algorithm for gradient-based optimization of stochastic objective functions. Our method is aimed towards machine learning problems with

算法 2: *AdaMax*, 基于无穷范数的 Adam 变体。详见 7.1 节。对于测试的机器学习问题, 良好默认设置为 $\alpha = 0.002$ 、 $\beta_1 = 0.9$ 和 $\beta_2 = 0.999$ 。用 β_1^t 表示 β_1 的 t 次幂。这里, $(\alpha/(1 - \beta_1^t))$ 是带有一阶矩偏差校正项的学习率。所有向量运算均为逐元素。

要求: α : 步长 **要求:** $\beta_1, \beta_2 \in [0, 1]$: 指数衰减率 **要求:** $f(\theta)$: 含参数 θ 的随机目标函数 **要求:** θ_0 : 初始参数向量 $m_0 \leftarrow 0$ (初始化 1st 矩向量) $u_0 \leftarrow 0$ (初始化指数加权无穷范数) $t \leftarrow 0$ (初始化时间步) **当** θ_t 未收敛 **时执行** $t \leftarrow t + 1$ $g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ (在时间步 t 上获取随机目标函数的梯度) $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (更新有偏一阶矩估计) $u_t \leftarrow \max(\beta_2 \cdot u_{t-1}, |g_t|)$ (更新指数加权无穷范数) $\theta_t \leftarrow \theta_{t-1} - (\alpha / (1 - \beta_1^t)) \cdot m_t / u_t$ (更新参数) **循环结束 返回** θ_t (所得参数)

请注意, 这里的衰减项等价地被参数化为 β_2^p 而非 β_{2^p} 。现在令 $p \rightarrow \infty$, 并定义 $u_t = \lim_{p \rightarrow \infty} (v_t)^{1/p}$, 则有:

$$u_t = \lim_{p \rightarrow \infty} (v_t)^{1/p} = \lim_{p \rightarrow \infty} \left((1 - \beta_2^p) \sum_{i=1}^t \beta_2^{p(t-i)} \cdot |g_i|^p \right)^{1/p} \quad (8)$$

$$= \lim_{p \rightarrow \infty} (1 - \beta_2^p)^{1/p} \left(\sum_{i=1}^t \beta_2^{p(t-i)} \cdot |g_i|^p \right)^{1/p} \quad (9)$$

$$= \lim_{p \rightarrow \infty} \left(\sum_{i=1}^t \left(\beta_2^{(t-i)} \cdot |g_i| \right)^p \right)^{1/p} \quad (10)$$

$$= \max(\beta_2^{t-1} |g_1|, \beta_2^{t-2} |g_2|, \dots, \beta_2 |g_{t-1}|, |g_t|) \quad (11)$$

这对应于一个异常简洁的递归公式:

$$u_t = \max(\beta_2 \cdot u_{t-1}, |g_t|) \quad (12)$$

其初始值为 $u_0 = 0$ 。值得注意的是, 方便之处在于, 在这种情况下我们无需校正初始化偏置。同样值得注意的是, 与 Adam 相比, AdaMax 中参数更新的幅度具有更简单的上界, 即: $|\Delta_t| \leq \alpha$ 。

7.2 时间平均

由于随机近似导致最后一次迭代带有噪声, 因此通常可以通过平均化获得更好的泛化性能。此前, Moulines 与 Bach (2011) 已证明 Polyak-Ruppert 平均 (Polyak & Juditsky, 1992; Ruppert, 1988) 可以改进标准 SGD 的收敛性, 其中 $\bar{\theta}_t = \frac{1}{t} \sum_{k=1}^t \theta_{k^0}$ 。另一种方法是使用参数的指数移动平均, 为较新的参数值赋予更高的权重。这可以很容易地实现, 只需在算法 1 和算法 2 的内层循环中添加一行代码: $\theta_t \leftarrow \beta_2 \cdot \theta_{t-1} + (1 - \beta_2) \theta_t$, 其中 $\theta_0 = 0$ 。初始化偏置同样可以通过估计器 $\hat{\theta}_t = \bar{\theta}_t / (1 - \beta_2^t)$ 进行校正。

8 结论

我们提出了一种简单且计算高效的算法, 用于随机目标函数的基于梯度的优化。我们的方法面向具有以下特征的机器学习问题:

large datasets and/or high-dimensional parameter spaces. The method combines the advantages of two recently popular optimization methods: the ability of AdaGrad to deal with sparse gradients, and the ability of RMSProp to deal with non-stationary objectives. The method is straightforward to implement and requires little memory. The experiments confirm the analysis on the rate of convergence in convex problems. Overall, we found Adam to be robust and well-suited to a wide range of non-convex optimization problems in the field machine learning.

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大规模数据集和/或高维参数空间。该方法结合了两种近期流行的优化方法的优点：AdaGrad处理稀疏梯度的能力，以及RMSProp处理非平稳目标的能力。该方法实现简单，且内存占用很小。实验证实了针对凸问题收敛率的分析。总体而言，我们发现Adam具有鲁棒性，非常适合机器学习领域中广泛的非凸优化问题。

9 致谢

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10 APPENDIX

10.1 CONVERGENCE PROOF

Definition 10.1. A function $f : R^d \rightarrow R$ is convex if for all $x, y \in R^d$, for all $\lambda \in [0, 1]$,

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(\lambda x + (1 - \lambda)y)$$

Also, notice that a convex function can be lower bounded by a hyperplane at its tangent.

Lemma 10.2. If a function $f : R^d \rightarrow R$ is convex, then for all $x, y \in R^d$,

$$f(y) \geq f(x) + \nabla f(x)^T(y - x)$$

The above lemma can be used to upper bound the regret and our proof for the main theorem is constructed by substituting the hyperplane with the Adam update rules.

The following two lemmas are used to support our main theorem. We also use some definitions simplify our notation, where $g_t \triangleq \nabla f_t(\theta_t)$ and $g_{t,i}$ as the i^{th} element. We define $g_{1:t,i} \in \mathbb{R}^t$ as a vector that contains the i^{th} dimension of the gradients over all iterations till t , $g_{1:t,i} = [g_{1,i}, g_{2,i}, \dots, g_{t,i}]$

Lemma 10.3. Let $g_t = \nabla f_t(\theta_t)$ and $g_{1:t}$ be defined as above and bounded, $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$. Then,

$$\sum_{t=1}^T \sqrt{\frac{g_{t,i}^2}{t}} \leq 2G_\infty \|g_{1:T,i}\|_2$$

Proof. We will prove the inequality using induction over T .

The base case for $T = 1$, we have $\sqrt{g_{1,i}^2} \leq 2G_\infty \|g_{1,i}\|_2$.

For the inductive step,

$$\begin{aligned} \sum_{t=1}^T \sqrt{\frac{g_{t,i}^2}{t}} &= \sum_{t=1}^{T-1} \sqrt{\frac{g_{t,i}^2}{t}} + \sqrt{\frac{g_{T,i}^2}{T}} \\ &\leq 2G_\infty \|g_{1:T-1,i}\|_2 + \sqrt{\frac{g_{T,i}^2}{T}} \\ &= 2G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} + \sqrt{\frac{g_{T,i}^2}{T}} \end{aligned}$$

From, $\|g_{1:T,i}\|_2^2 - g_{T,i}^2 + \frac{g_{T,i}^4}{4\|g_{1:T,i}\|_2^2} \geq \|g_{1:T,i}\|_2^2 - g_{T,i}^2$, we can take square root of both side and have,

$$\begin{aligned} \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\|g_{1:T,i}\|_2} \\ &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\sqrt{TG_\infty^2}} \end{aligned}$$

Rearrange the inequality and substitute the $\sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2}$ term,

$$G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} + \sqrt{\frac{g_{T,i}^2}{T}} \leq 2G_\infty \|g_{1:T,i}\|_2$$

□

10 附录

10.1 收敛证明

定义 10.1. 若对所有 $x, y \in R^d$, 以及所有 $\lambda \in [0, 1]$, 函数 $f: R^d \rightarrow R$ 满足凸函数条件, 则称其为凸函数,

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(\lambda x + (1 - \lambda)y)$$

同时, 请注意凸函数在其切平面处可以被一个超平面从下界约束。

引理 10.2. 若函数 $f: R^d \rightarrow R$ 为凸函数, 则对所有 $x, y \in R^d$,

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

上述引理可用于为遗憾值建立上界, 我们对主定理的证明是通过用Adam更新规则替换超平面来构建的。

以下两个引理用于支撑我们的主定理。我们还使用一些定义来简化记号, 其中 $g_t \triangleq \nabla f_t(\theta_t)$ 和 $g_{t,i}$ 作为 i^{th} 元素。我们定义 $g_{1:t,i} \in \mathbb{R}^t$ 为一个向量, 它包含截至 t 的所有迭代中梯度的 i^{th} 维度, $g_{1:t,i} = [g_{1,i}, g_{2,i}, \dots, g_{t,i}]$

Le引理 10.3. 设 $g_t = \nabla f_t(\theta_t)$ 和 $g_{1:t}$ 如上定义且有界, $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$. 那么,

$$\sum_{t=1}^T \sqrt{\frac{g_{t,i}^2}{t}} \leq 2G_\infty \|g_{1:T,i}\|_2$$

证明。我们将对T使用归纳法来证明该不等式。

对于 $T = 1$ 的基础情形, 我们有 $\sqrt{g_{1,i}^2} \leq 2G_\infty \|g_{1,i}\|_2$ 。

对于归纳步骤,

$$\begin{aligned} \sum_{t=1}^T \sqrt{\frac{g_{t,i}^2}{t}} &= \sum_{t=1}^{T-1} \sqrt{\frac{g_{t,i}^2}{t}} + \sqrt{\frac{g_{T,i}^2}{T}} \\ &\leq 2G_\infty \|g_{1:T-1,i}\|_2 + \sqrt{\frac{g_{T,i}^2}{T}} \\ &= 2G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} + \sqrt{\frac{g_{T,i}^2}{T}} \end{aligned}$$

由, $\|g_{1:T,i}\|_2^2 - g_{T,i}^2 + \frac{g_{T,i}^4}{4\|g_{1:T,i}\|_2^2} \geq \|g_{1:T,i}\|_2^2 - g_{T,i}^2$, 我们可以对等式两边取平方根, 得到,

$$\begin{aligned} \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\|g_{1:T,i}\|_2} \\ &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\sqrt{T}G_\infty^2} \end{aligned}$$

重新整理该不等式, 并代入 $\sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2}$ 项, 得到

$$G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} + \sqrt{\frac{g_{T,i}^2}{T}} \leq 2G_\infty \|g_{1:T,i}\|_2$$

□

Lemma 10.4. Let $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. For $\beta_1, \beta_2 \in [0, 1)$ that satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$ and bounded $g_t, \|g_t\|_2 \leq G, \|g_t\|_\infty \leq G_\infty$, the following inequality holds

$$\sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} \leq \frac{2}{1-\gamma} \frac{1}{\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

Proof. Under the assumption, $\frac{\sqrt{1-\beta_2^t}}{(1-\beta_1^t)^2} \leq \frac{1}{(1-\beta_1)^2}$. We can expand the last term in the summation using the update rules in Algorithm 1,

$$\begin{aligned} \sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} &= \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(\sum_{k=1}^T (1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T \sum_{j=1}^T (1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^T \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T \sum_{j=1}^T (1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^T \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T(1-\beta_2)\beta_2^{T-k} g_{k,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(1-\beta_1)^2}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^T T \left(\frac{\beta_1^2}{\sqrt{\beta_2}} \right)^{T-k} \|g_{k,i}\|_2 \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{T}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^T \gamma^{T-k} \|g_{k,i}\|_2 \end{aligned}$$

Similarly, we can upper bound the rest of the terms in the summation.

$$\begin{aligned} \sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} &\leq \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T-t} t\gamma^j \\ &\leq \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^T t\gamma^j \end{aligned}$$

For $\gamma < 1$, using the upper bound on the arithmetic-geometric series, $\sum_t t\gamma^t < \frac{1}{(1-\gamma)^2}$:

$$\sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^T t\gamma^j \leq \frac{1}{(1-\gamma)^2 \sqrt{1-\beta_2}} \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t}}$$

Apply Lemma 10.3,

$$\sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} \leq \frac{2G_\infty}{(1-\gamma)^2 \sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

□

To simplify the notation, we define $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. Intuitively, our following theorem holds when the learning rate α_t is decaying at a rate of $t^{-\frac{1}{2}}$ and first moment running average coefficient $\beta_{1,t}$ decay exponentially with λ , that is typically close to 1, e.g. $1 - 10^{-8}$.

Theorem 10.5. Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G, \|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in \mathbb{R}^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$,

引理 10.4. 设 $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$ 对于满足 $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$ 且有界的 g_t $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$ 的 $\beta_1, \beta_2 \in [0, 1)$, 以下不等式成立

$$\sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} \leq \frac{2}{1-\gamma} \frac{1}{\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

证明。在该假设下, $\frac{\sqrt{1-\beta_2^t}}{(1-\beta_1^t)^2} \leq \frac{1}{(1-\beta_1)^2}$ 。我们可以利用算法 1 中的更新规则展开求和式中的最后一项,

$$\begin{aligned} \sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} &= \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(\sum_{k=1}^T (1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T \sum_{j=1}^T (1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^T \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T \sum_{j=1}^T (1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^T \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T(1-\beta_2)\beta_2^{T-k} g_{k,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(1-\beta_1)^2}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^T T \left(\frac{\beta_1^2}{\sqrt{\beta_2}} \right)^{T-k} \|g_{k,i}\|_2 \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{T}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^T \gamma^{T-k} \|g_{k,i}\|_2 \end{aligned}$$

类似地, 我们可以对求和式中的其余各项求上界。

$$\begin{aligned} \sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} &\leq \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T-t} t\gamma^j \\ &\leq \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^T t\gamma^j \end{aligned}$$

对于 $\gamma < 1$, 利用算术-几何级数的上界, $\sum_t t\gamma^t < \frac{1}{(1-\gamma)^2}$:

$$\sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^T t\gamma^j \leq \frac{1}{(1-\gamma)^2 \sqrt{1-\beta_2}} \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t}}$$

应用引理 10.3,

$$\sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} \leq \frac{2G_\infty}{(1-\gamma)^2 \sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

□

为简化记号, 我们定义 $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$ 。直观地说, 当学习率 α_t 以 $t^{-\frac{1}{2}}$ 的速率衰减, 且一阶矩滑动平均系数 β_1 随 t 指数衰减 (λ 通常接近 1, 例如 $1 - 10^{-8}$) 时, 以下定理成立。

定理 10.5 f_t θ_t $\|\nabla f_t \theta\|_2 \leq G$ $\|\nabla f_t \theta\|_\infty \leq \dots$ 假设该函数具有有界梯度, G_∞ $\theta \in R^d$ $\|\theta_n - \theta_m\|_2 \leq D$ $(\cdot, \cdot)$ 对于所有成立, 且 Adam 生成的任意之间的距离有界, ,

$\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, \dots, T\}$, and $\beta_1, \beta_2 \in [0, 1)$ satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$. Let $\alpha_t = \frac{\alpha}{\sqrt{t}}$ and $\beta_{1,t} = \beta_1 \lambda^{t-1}$, $\lambda \in (0, 1)$. Adam achieves the following guarantee, for all $T \geq 1$.

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(\beta_1+1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \sum_{i=1}^d \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Proof. Using Lemma 10.2, we have,

$$f_t(\theta_t) - f_t(\theta^*) \leq g_t^T(\theta_t - \theta^*) = \sum_{i=1}^d g_{t,i}(\theta_{t,i} - \theta_{*,i}^*)$$

From the update rules presented in algorithm 1,

$$\begin{aligned} \theta_{t+1} &= \theta_t - \alpha_t \widehat{m}_t / \sqrt{\widehat{v}_t} \\ &= \theta_t - \frac{\alpha_t}{1-\beta_1^t} \left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_t}} m_{t-1} + \frac{(1-\beta_{1,t})}{\sqrt{\widehat{v}_t}} g_t \right) \end{aligned}$$

We focus on the i^{th} dimension of the parameter vector $\theta_t \in R^d$. Subtract the scalar $\theta_{*,i}^*$ and square both sides of the above update rule, we have,

$$(\theta_{t+1,i} - \theta_{*,i}^*)^2 = (\theta_{t,i} - \theta_{*,i}^*)^2 - \frac{2\alpha_t}{1-\beta_1^t} \left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} m_{t-1,i} + (1 - \frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} g_{t,i}) \right) (\theta_{t,i} - \theta_{*,i}^*) + \alpha_t^2 \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}} \right)^2$$

We can rearrange the above equation and use Young's inequality, $ab \leq a^2/2 + b^2/2$. Also, it can be shown that $\sqrt{\widehat{v}_{t,i}} = \sqrt{\sum_{j=1}^t (1-\beta_2)\beta_2^{t-j} g_{j,i}^2} / \sqrt{1-\beta_2^t} \leq \|g_{1:t,i}\|_2$ and $\beta_{1,t} \leq \beta_1$. Then

$$\begin{aligned} g_{t,i}(\theta_{t,i} - \theta_{*,i}^*) &= \frac{(1-\beta_1^t)\sqrt{\widehat{v}_{t,i}}}{2\alpha_t(1-\beta_{1,t})} \left((\theta_{t,i} - \theta_{*,i}^*)^2 - (\theta_{t+1,i} - \theta_{*,i}^*)^2 \right) \\ &\quad + \frac{\beta_{1,t}}{(1-\beta_{1,t})} \frac{\widehat{v}_{t-1,i}^{\frac{1}{4}}}{\sqrt{\alpha_{t-1}}} (\theta_{*,i}^* - \theta_{t,i}) \sqrt{\alpha_{t-1}} \frac{m_{t-1,i}}{\widehat{v}_{t-1,i}^{\frac{1}{4}}} + \frac{\alpha_t(1-\beta_1^t)\sqrt{\widehat{v}_{t,i}}}{2(1-\beta_{1,t})} \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}} \right)^2 \\ &\leq \frac{1}{2\alpha_t(1-\beta_1)} \left((\theta_{t,i} - \theta_{*,i}^*)^2 - (\theta_{t+1,i} - \theta_{*,i}^*)^2 \right) \sqrt{\widehat{v}_{t,i}} + \frac{\beta_{1,t}}{2\alpha_{t-1}(1-\beta_{1,t})} (\theta_{*,i}^* - \theta_{t,i})^2 \sqrt{\widehat{v}_{t-1,i}} \\ &\quad + \frac{\beta_1 \alpha_{t-1}}{2(1-\beta_1)} \frac{m_{t-1,i}^2}{\sqrt{\widehat{v}_{t-1,i}}} + \frac{\alpha_t}{2(1-\beta_1)} \frac{\widehat{m}_{t,i}^2}{\sqrt{\widehat{v}_{t,i}}} \end{aligned}$$

We apply Lemma 10.4 to the above inequality and derive the regret bound by summing across all the dimensions for $i \in 1, \dots, d$ in the upper bound of $f_t(\theta_t) - f_t(\theta^*)$ and the sequence of convex functions for $t \in 1, \dots, T$:

$$\begin{aligned} R(T) &\leq \sum_{i=1}^d \frac{1}{2\alpha_1(1-\beta_1)} (\theta_{1,i} - \theta_{*,i}^*)^2 \sqrt{\widehat{v}_{1,i}} + \sum_{i=1}^d \sum_{t=2}^T \frac{1}{2(1-\beta_1)} (\theta_{t,i} - \theta_{*,i}^*)^2 \left(\frac{\sqrt{\widehat{v}_{t,i}}}{\alpha_t} - \frac{\sqrt{\widehat{v}_{t-1,i}}}{\alpha_{t-1}} \right) \\ &\quad + \frac{\beta_1 \alpha G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \frac{\alpha G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 \\ &\quad + \sum_{i=1}^d \sum_{t=1}^T \frac{\beta_{1,t}}{2\alpha_t(1-\beta_{1,t})} (\theta_{*,i}^* - \theta_{t,i})^2 \sqrt{\widehat{v}_{t,i}} \end{aligned}$$

$\|\theta_m - \theta_n\|_\infty \leq D_\infty$ 对于任意 $m, n \in \{1, \dots, T\}$, 以及 $\beta_1, \beta_2 \in [0, 1)$ 满足 $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$ 。令 $\alpha_t = \frac{\alpha}{\sqrt{t}}$ 和 $\beta_{1,t} = \beta_1 \lambda^{t-1}$, $\lambda \in (0, 1)$ 。对于所有 $T \geq 1$, Adam 可实现如下保证。

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(\beta_1+1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \sum_{i=1}^d \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

证明。由引理 10.2, 我们有,

$$f_t(\theta_t) - f_t(\theta^*) \leq g_t^T(\theta_t - \theta^*) = \sum_{i=1}^d g_{t,i}(\theta_{t,i} - \theta_{*,i}^*)$$

根据算法 1 中给出的更新规则,

$$\begin{aligned} \theta_{t+1} &= \theta_t - \alpha_t \widehat{m}_t / \sqrt{\widehat{v}_t} \\ &= \theta_t - \frac{\alpha_t}{1-\beta_1^t} \left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_t}} m_{t-1} + \frac{(1-\beta_{1,t})}{\sqrt{\widehat{v}_t}} g_t \right) \end{aligned}$$

我们关注参数向量 $\theta_t \in R^d$ 的第 i^{th} 维。对标量 $\theta_{*,i}^*$ 作差, 并将上述更新规则两边平方, 我们有,

$$(\theta_{t+1,i} - \theta_{*,i}^*)^2 = (\theta_{t,i} - \theta_{*,i}^*)^2 - \frac{2\alpha_t}{1-\beta_1^t} \left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} m_{t-1,i} + (1 - \frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} g_{t,i}) \right) (\theta_{t,i} - \theta_{*,i}^*) + \alpha_t^2 \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}} \right)^2$$

我们可以重排上述方程并应用杨氏不等式, $ab \leq a^2/2 + b^2/2$ 。此外, 可以证明 $\sqrt{\widehat{v}_t}$, $i = \sqrt{\sum_{j=1}^t (1-\beta_2)\beta_2^{t-j} g_{j,i}^2 / \sqrt{1-\beta_2}} \leq \|g_{1:t,i}\|_2$ 以及 $\beta_{1,t} \leq \beta_{1,0}$ 。于是

$$\begin{aligned} g_{t,i}(\theta_{t,i} - \theta_{*,i}^*) &= \frac{(1-\beta_1^t)\sqrt{\widehat{v}_{t,i}}}{2\alpha_t(1-\beta_{1,t})} \left((\theta_{t,i} - \theta_{*,i}^*)^2 - (\theta_{t+1,i} - \theta_{*,i}^*)^2 \right) \\ &\quad + \frac{\beta_{1,t}}{(1-\beta_{1,t})} \frac{\widehat{v}_{t-1,i}^{\frac{1}{4}}}{\sqrt{\alpha_{t-1}}} (\theta_{*,i}^* - \theta_{t,i}) \sqrt{\alpha_{t-1}} \frac{m_{t-1,i}}{\widehat{v}_{t-1,i}^{\frac{1}{4}}} + \frac{\alpha_t(1-\beta_1^t)\sqrt{\widehat{v}_{t,i}}}{2(1-\beta_{1,t})} \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}} \right)^2 \\ &\leq \frac{1}{2\alpha_t(1-\beta_1)} \left((\theta_{t,i} - \theta_{*,i}^*)^2 - (\theta_{t+1,i} - \theta_{*,i}^*)^2 \right) \sqrt{\widehat{v}_{t,i}} + \frac{\beta_{1,t}}{2\alpha_{t-1}(1-\beta_{1,t})} (\theta_{*,i}^* - \theta_{t,i})^2 \sqrt{\widehat{v}_{t-1,i}} \\ &\quad + \frac{\beta_1 \alpha_{t-1}}{2(1-\beta_1)} \frac{m_{t-1,i}^2}{\sqrt{\widehat{v}_{t-1,i}}} + \frac{\alpha_t}{2(1-\beta_1)} \frac{\widehat{m}_{t,i}^2}{\sqrt{\widehat{v}_{t,i}}} \end{aligned}$$

我们将引理 10.4 应用于上述不等式, 并对 $f_t(\theta_t) - f_t(\theta^*)$ 上界中所有维度 $i \in 1, \dots, d$ 以及凸函数序列 $t \in 1, \dots, T$ 求和, 推导出遗憾界:

$$\begin{aligned} R(T) &\leq \sum_{i=1}^d \frac{1}{2\alpha_1(1-\beta_1)} (\theta_{1,i} - \theta_{*,i}^*)^2 \sqrt{\widehat{v}_{1,i}} + \sum_{i=1}^d \sum_{t=2}^T \frac{1}{2(1-\beta_1)} (\theta_{t,i} - \theta_{*,i}^*)^2 \left(\frac{\sqrt{\widehat{v}_{t,i}}}{\alpha_t} - \frac{\sqrt{\widehat{v}_{t-1,i}}}{\alpha_{t-1}} \right) \\ &\quad + \frac{\beta_1 \alpha G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \frac{\alpha G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 \\ &\quad + \sum_{i=1}^d \sum_{t=1}^T \frac{\beta_{1,t}}{2\alpha_t(1-\beta_{1,t})} (\theta_{*,i}^* - \theta_{t,i})^2 \sqrt{\widehat{v}_{t,i}} \end{aligned}$$

From the assumption, $\|\theta_t - \theta^*\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$, we have:

$$\begin{aligned}
R(T) &\leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \frac{D_\infty^2}{2\alpha} \sum_{i=1}^d \sum_{t=1}^t \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t\widehat{v}_{t,i}} \\
&\leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 \\
&\quad + \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha} \sum_{i=1}^d \sum_{t=1}^t \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t}
\end{aligned}$$

We can use arithmetic geometric series upper bound for the last term:

$$\begin{aligned}
\sum_{t=1}^t \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t} &\leq \sum_{t=1}^t \frac{1}{(1-\beta_1)} \lambda^{t-1} \sqrt{t} \\
&\leq \sum_{t=1}^t \frac{1}{(1-\beta_1)} \lambda^{t-1} t \\
&\leq \frac{1}{(1-\beta_1)(1-\lambda)^2}
\end{aligned}$$

Therefore, we have the following regret bound:

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \sum_{i=1}^d \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha\beta_1(1-\lambda)^2}$$

□

根据假设, $\|\theta_t - \theta^*\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$, 我们有:

$$\begin{aligned}
 R(T) &\leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \frac{D_\infty^2}{2\alpha} \sum_{i=1}^d \sum_{t=1}^t \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t\widehat{v}_{t,i}} \\
 &\leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 \\
 &\quad + \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha} \sum_{i=1}^d \sum_{t=1}^t \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t}
 \end{aligned}$$

我们可以对最后一项使用算术几何级数上界:

$$\begin{aligned}
 \sum_{t=1}^t \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t} &\leq \sum_{t=1}^t \frac{1}{(1-\beta_1)} \lambda^{t-1} \sqrt{t} \\
 &\leq \sum_{t=1}^t \frac{1}{(1-\beta_1)} \lambda^{t-1} t \\
 &\leq \frac{1}{(1-\beta_1)(1-\lambda)^2}
 \end{aligned}$$

因此, 我们得到如下遗憾界:

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \sum_{i=1}^d \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha\beta_1(1-\lambda)^2}$$

□